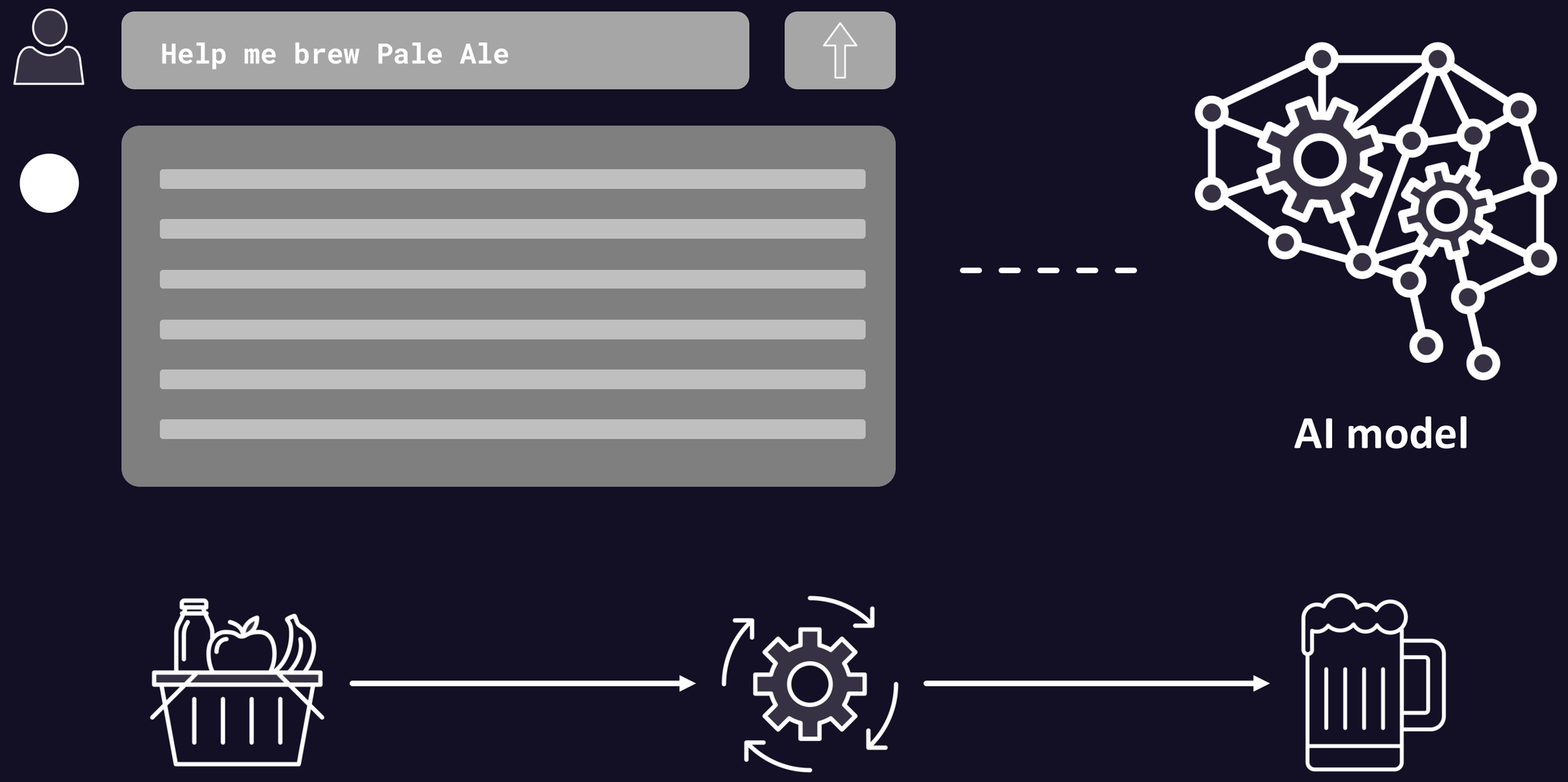
Applying Multi-agent Systems to Daily Tasks

Using Multi-agent Systems

Bogdan Sucaciu

Engineering Lead

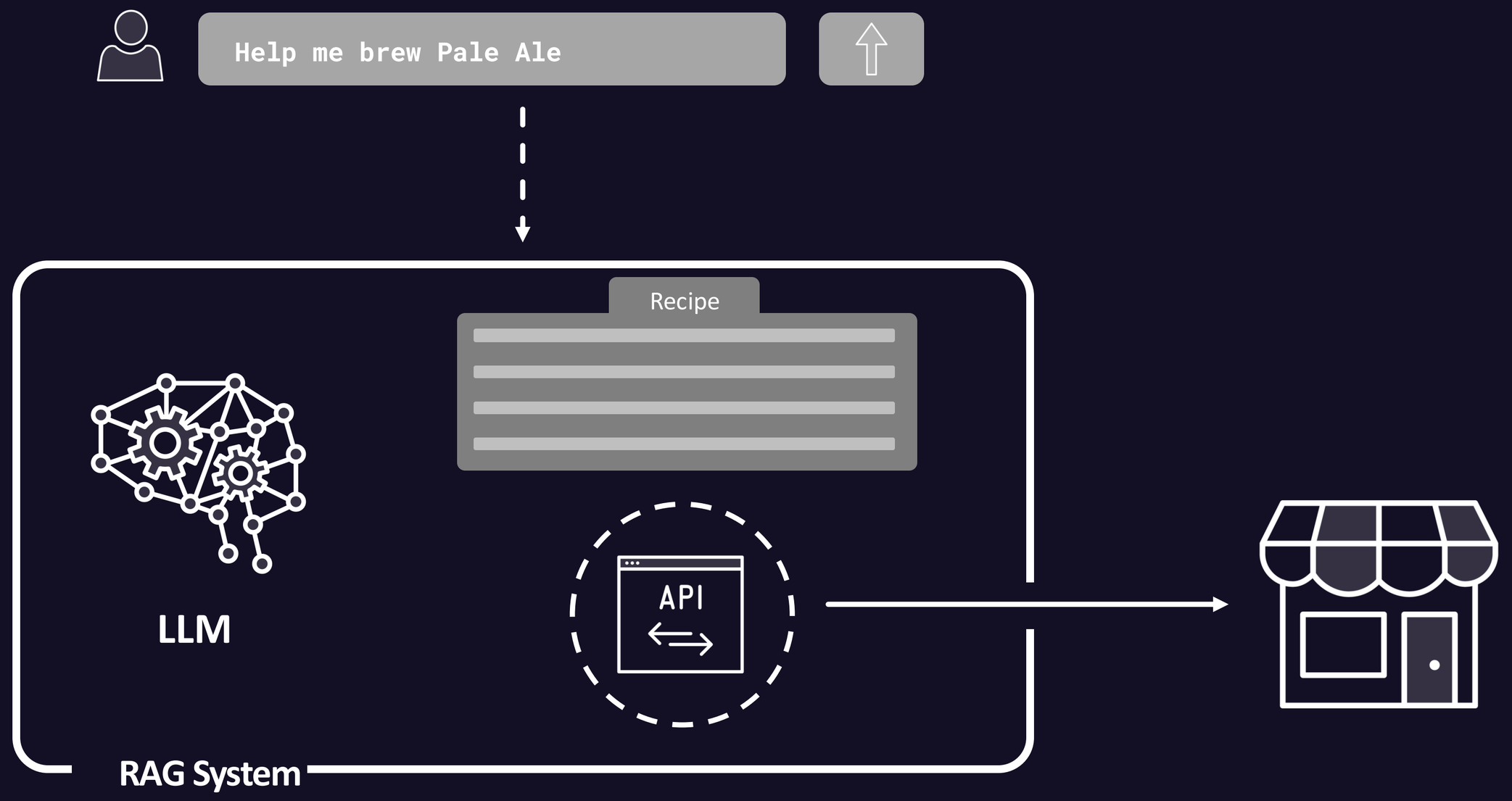
LinkedIn: Bogdan Sucaciu

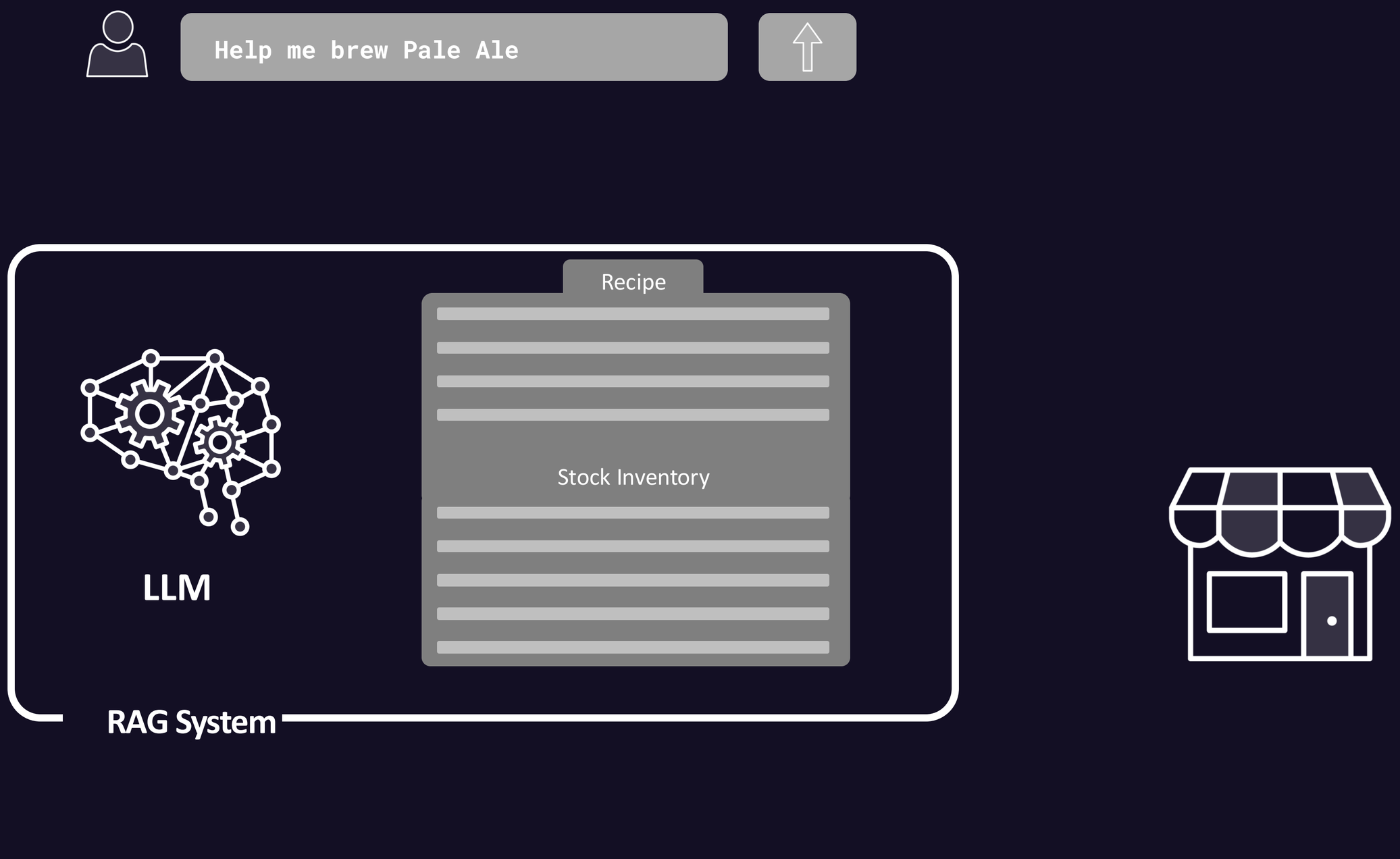

Help me brew Pale Ale
AI model

Retrieval-Augmented Generation (RAG)

“A technique that grants generative AI models information retrieval capabilities.”

Citation: wikipedia.org/wiki/Retrieval-augmented_generation


Help me brew Pale Ale
Recipe
LLM
API
RAG System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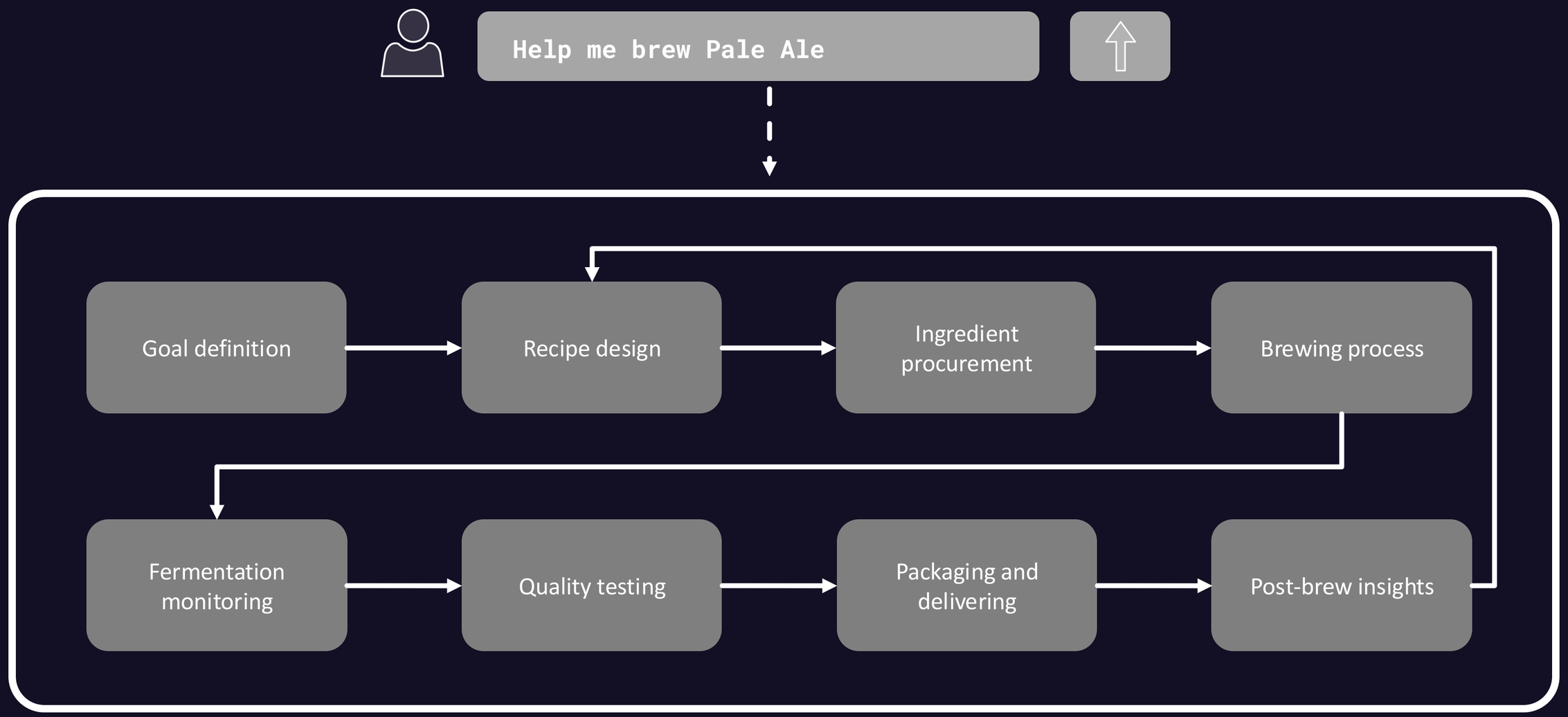
Recipe
Stock Inventory
LLM
RAG System


Help me brew Pale Ale
Goal definition
Recipe design
Ingredient procurement
Brewing process
Fermentation monitoring
Quality testing
Packaging and delivering
Post-brew insights

Agentic AI System

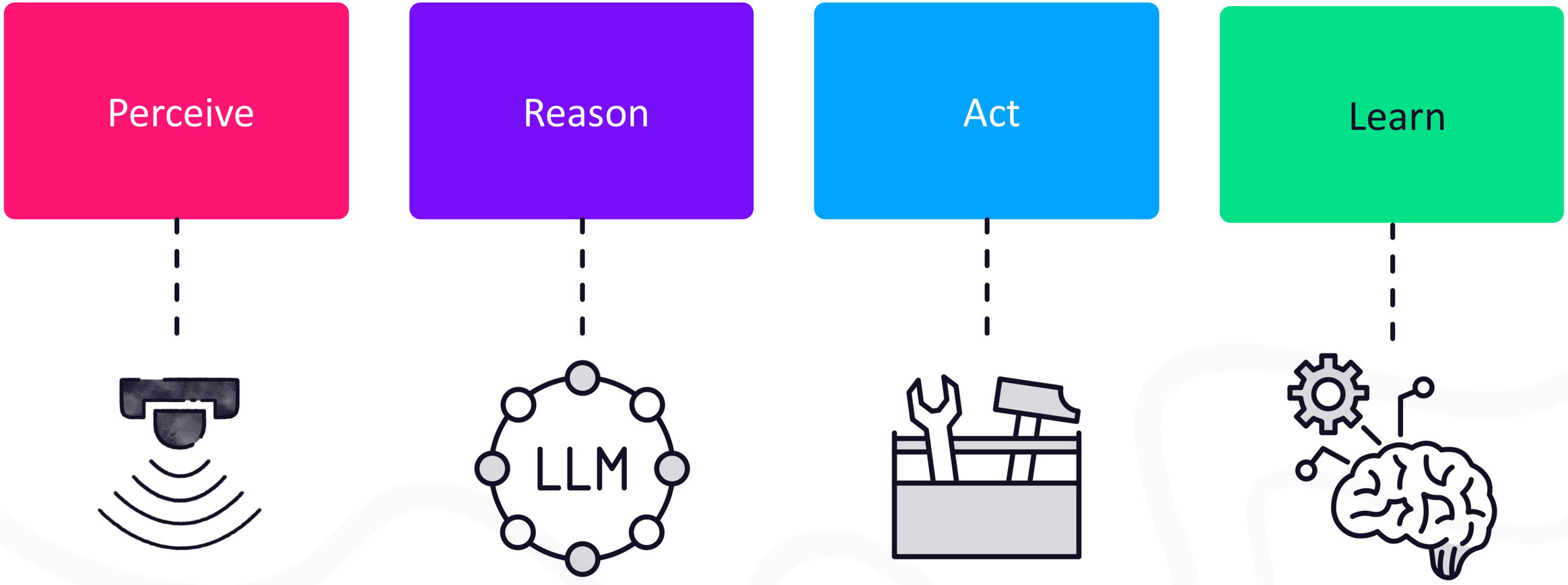
Autonomous execution

Proactivity

Goal-oriented

Collaboration

Agentic AI



Agentic AI

Perceive

Reason

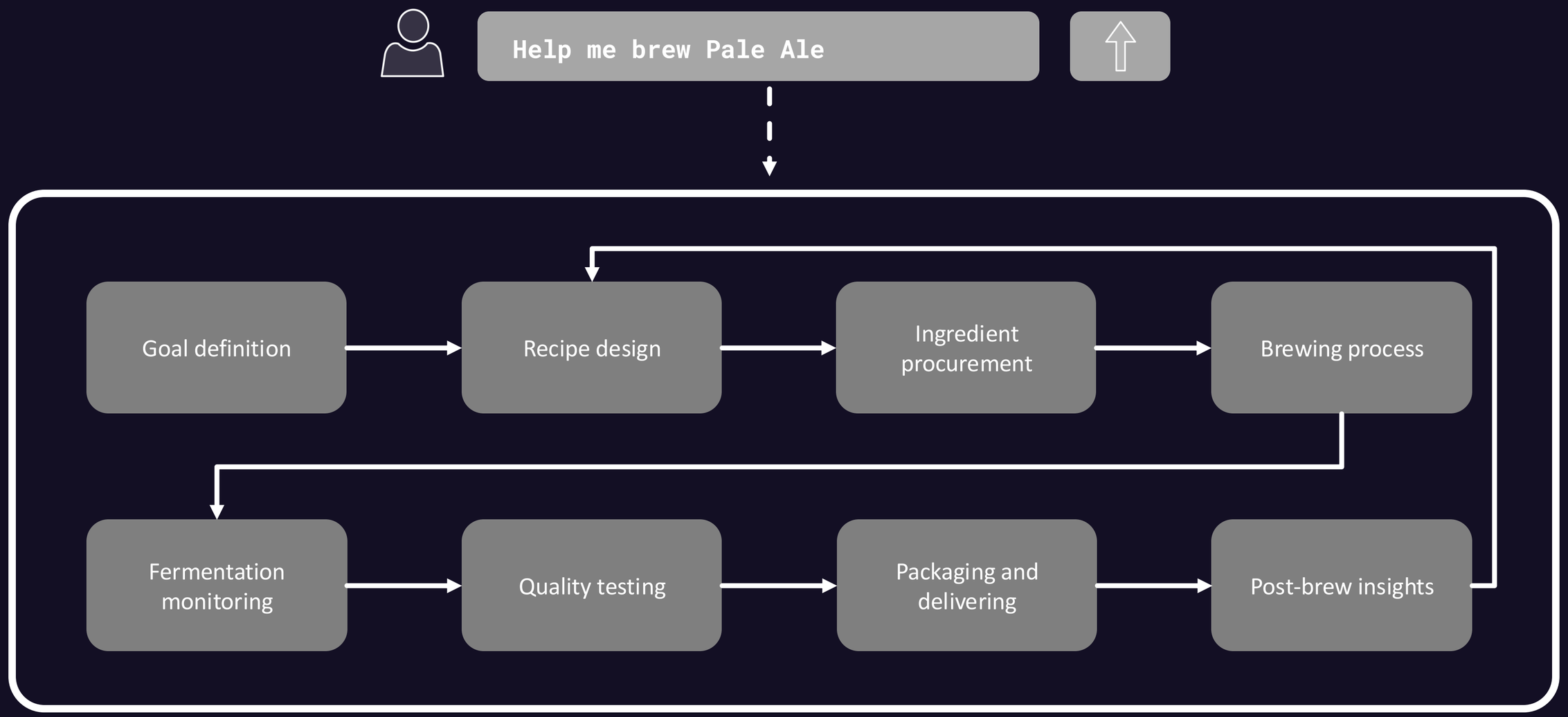
Act

Learn

Plan

Act

Adapt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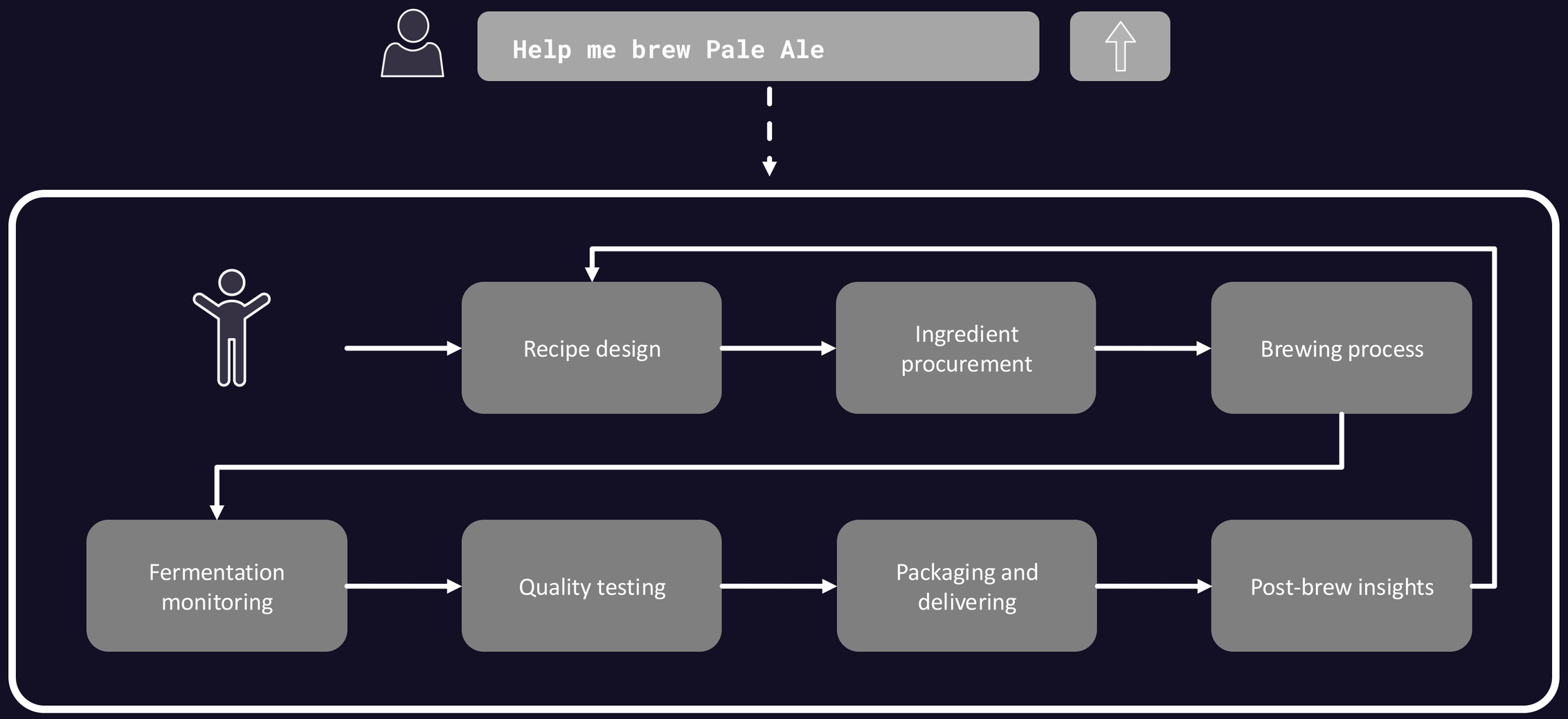
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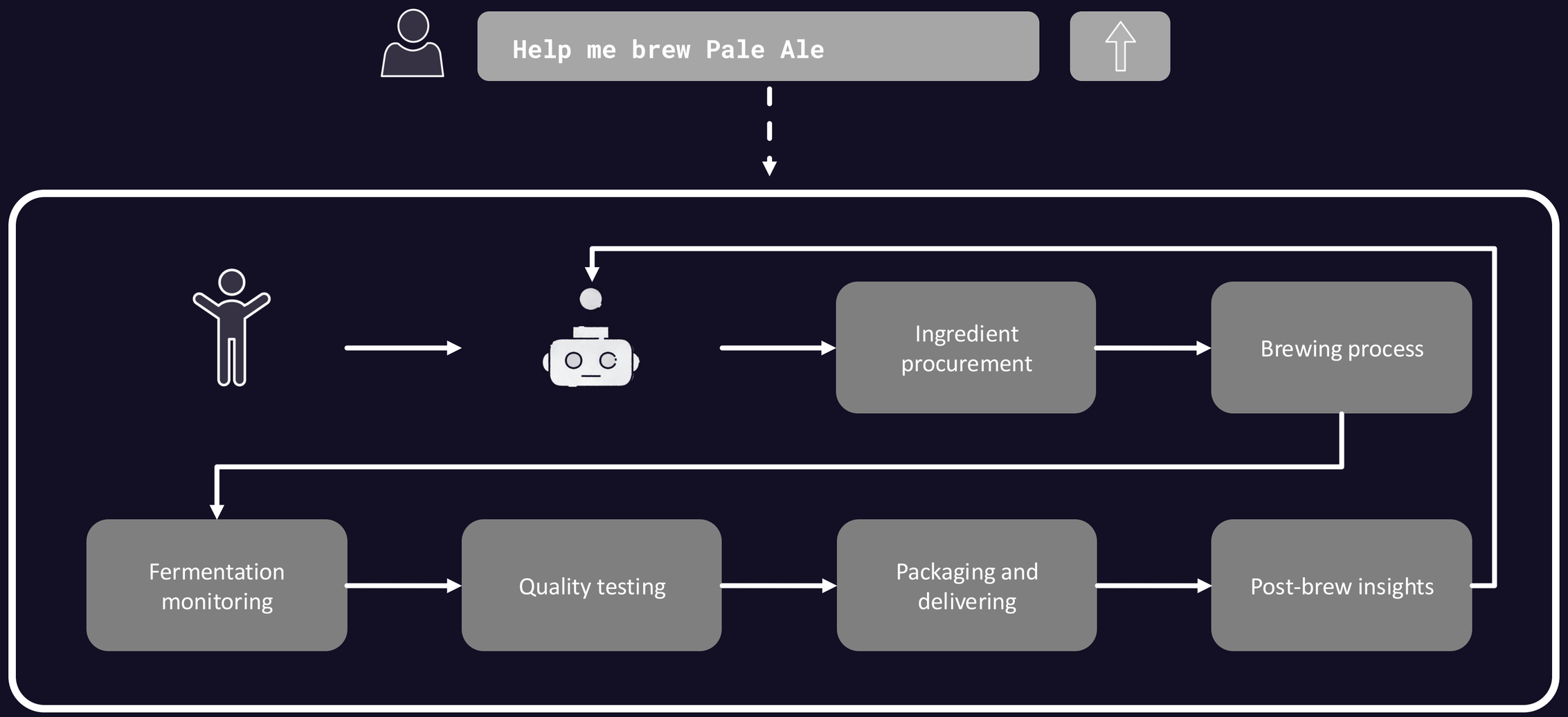
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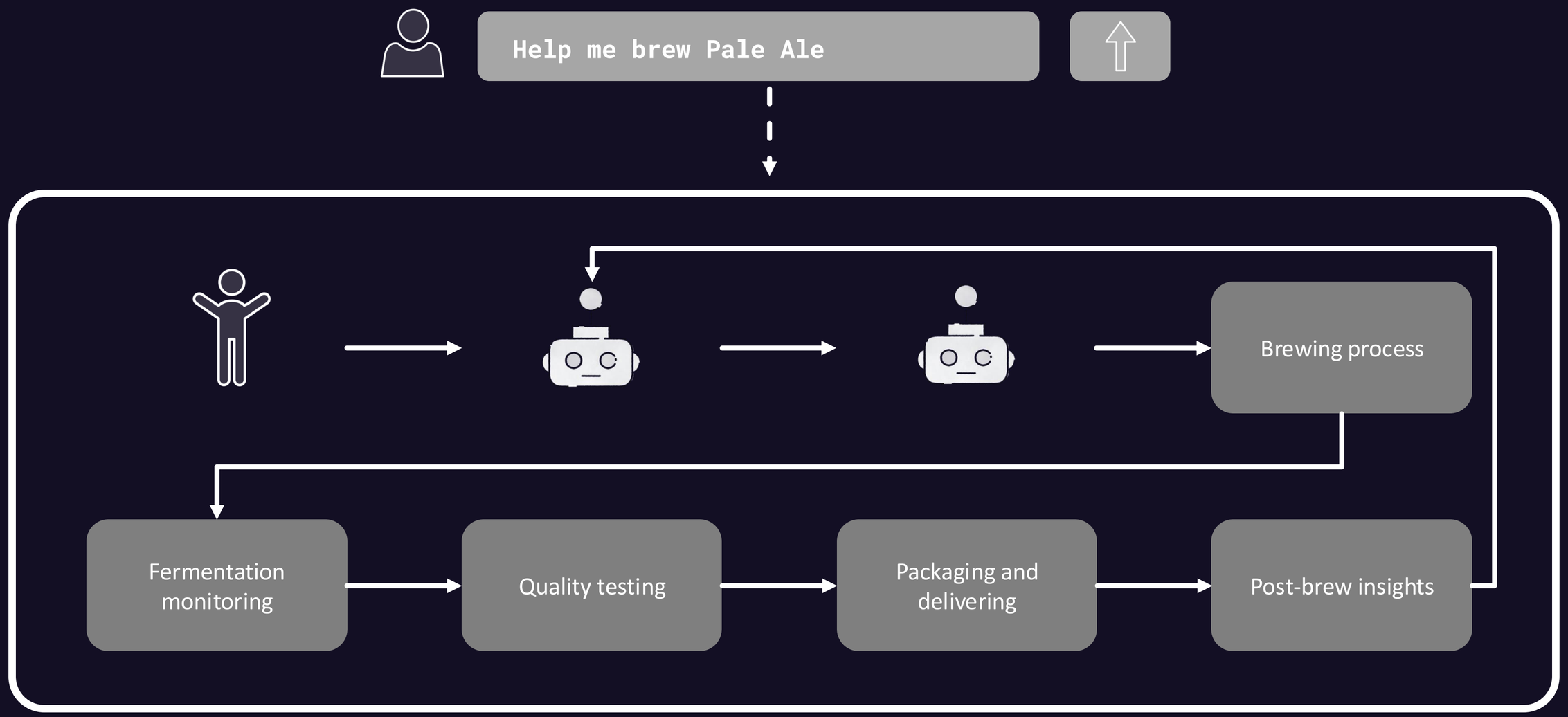
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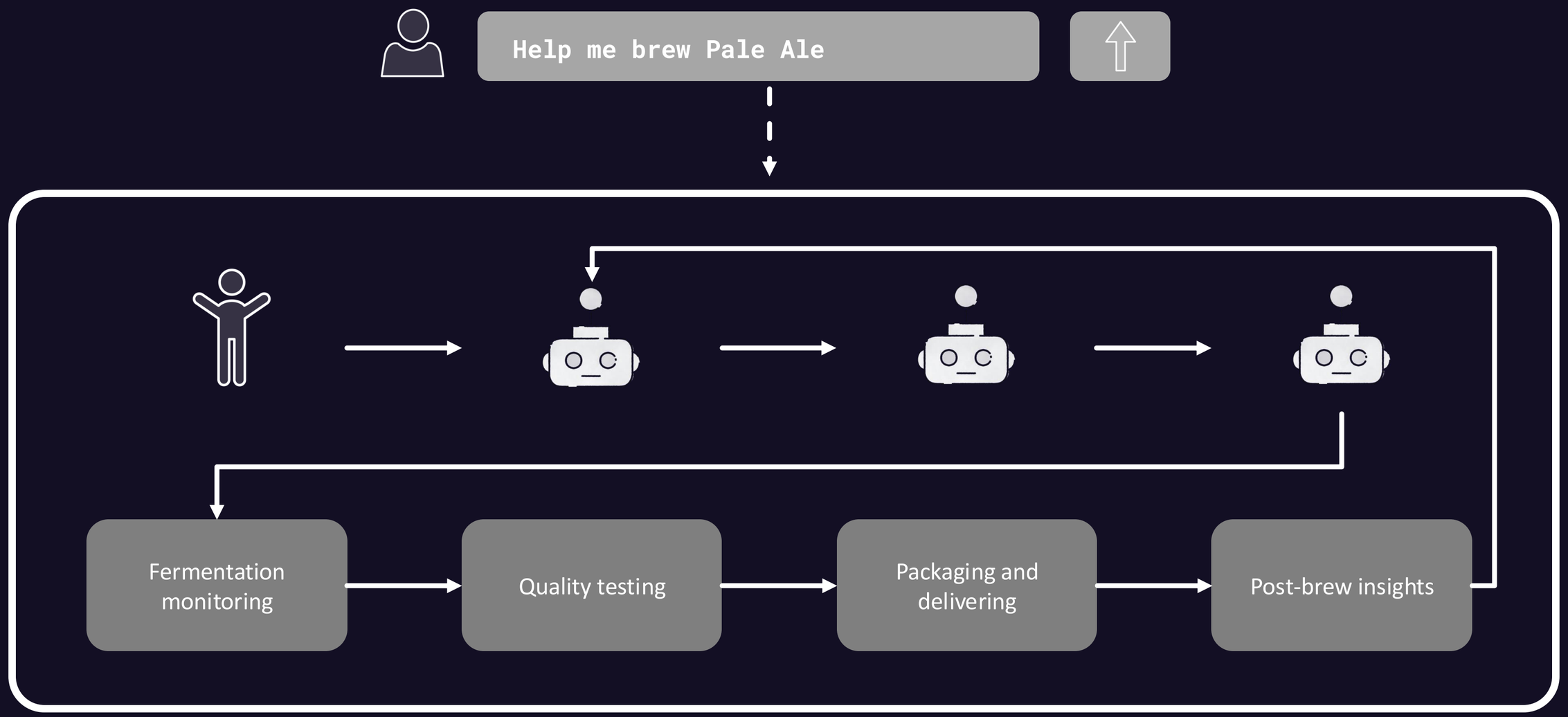
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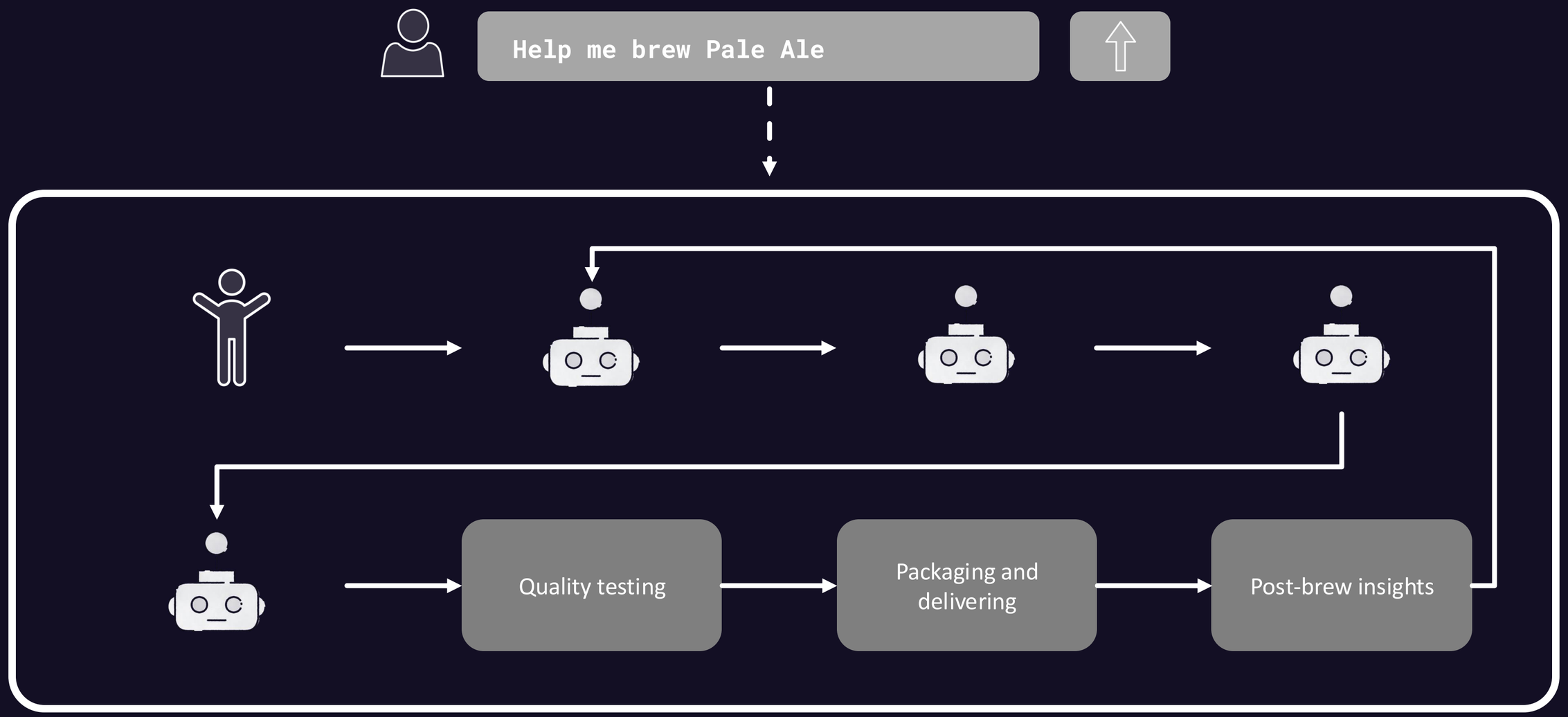
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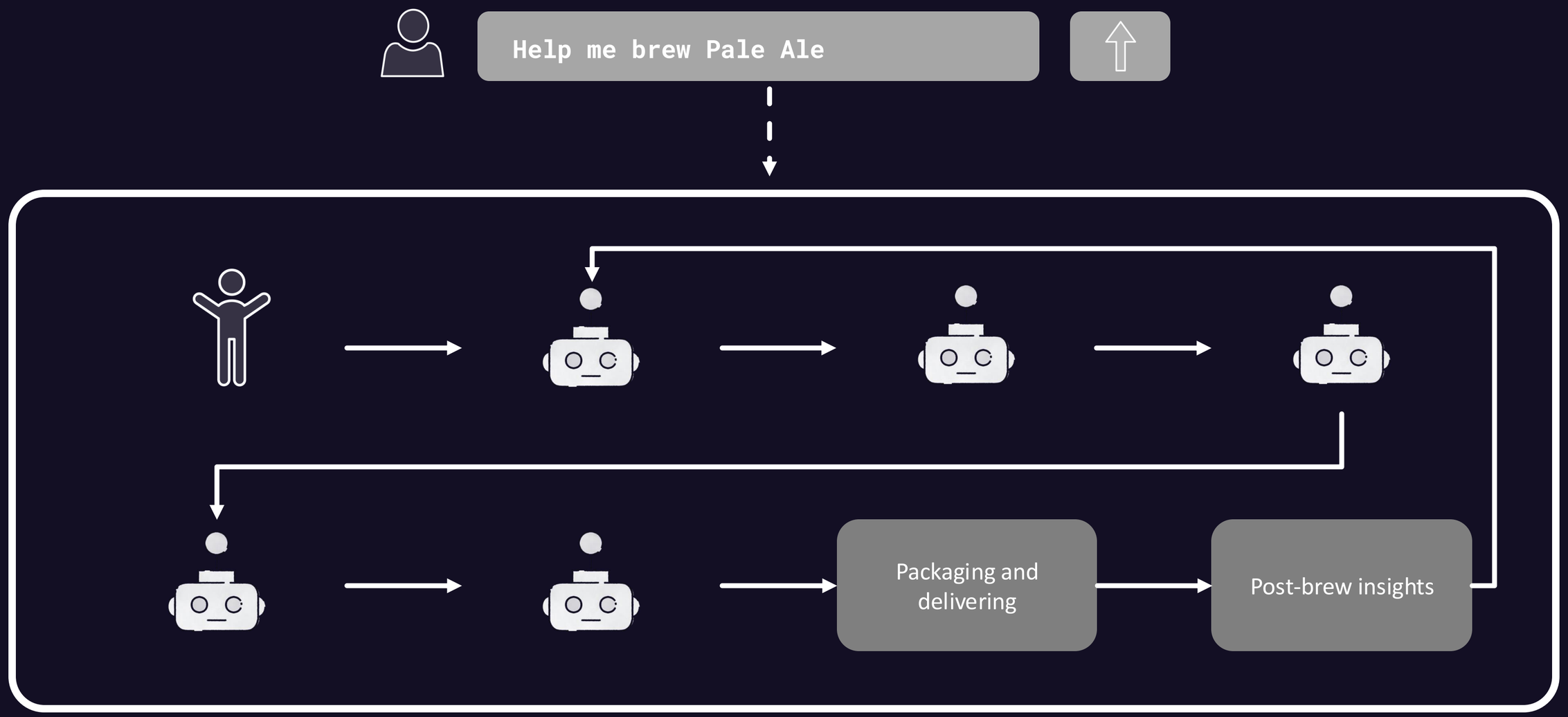
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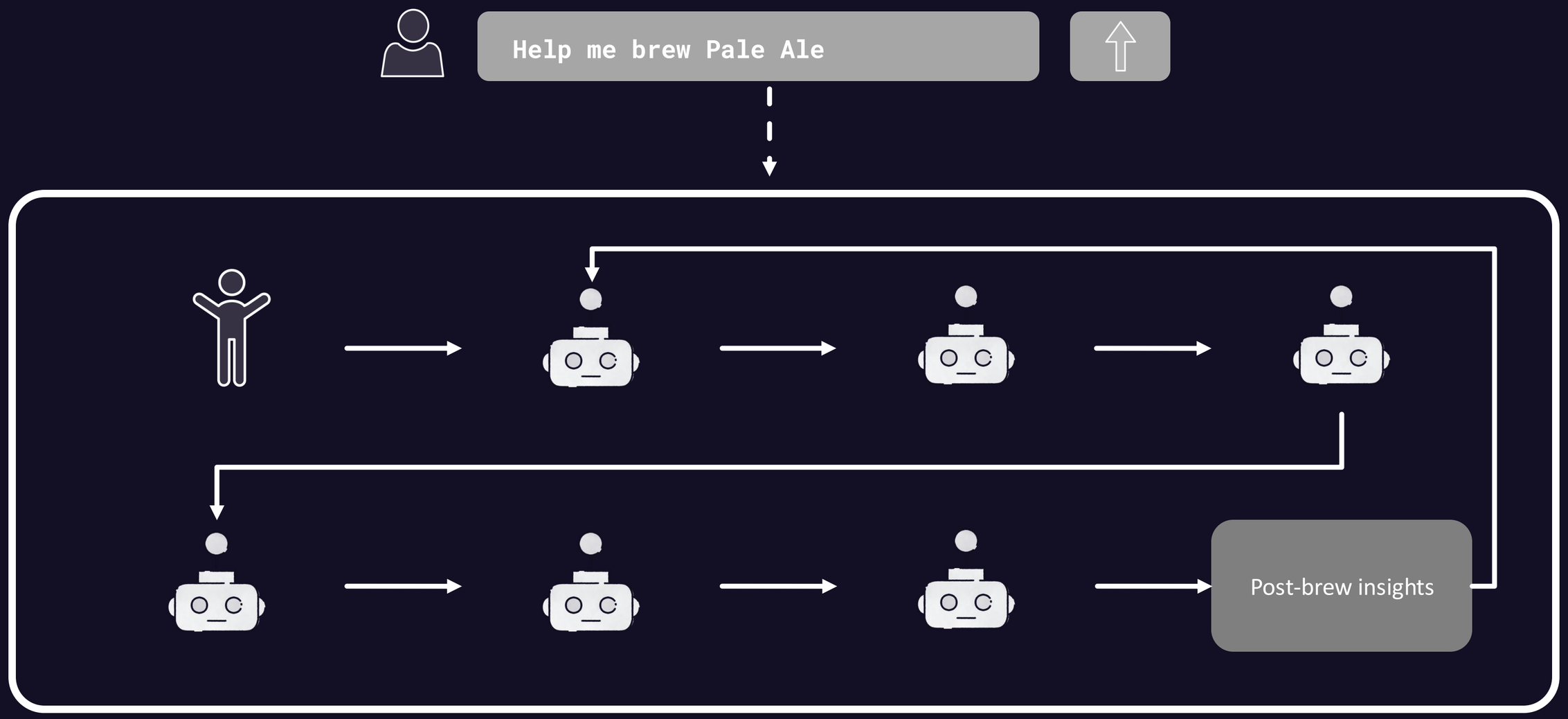
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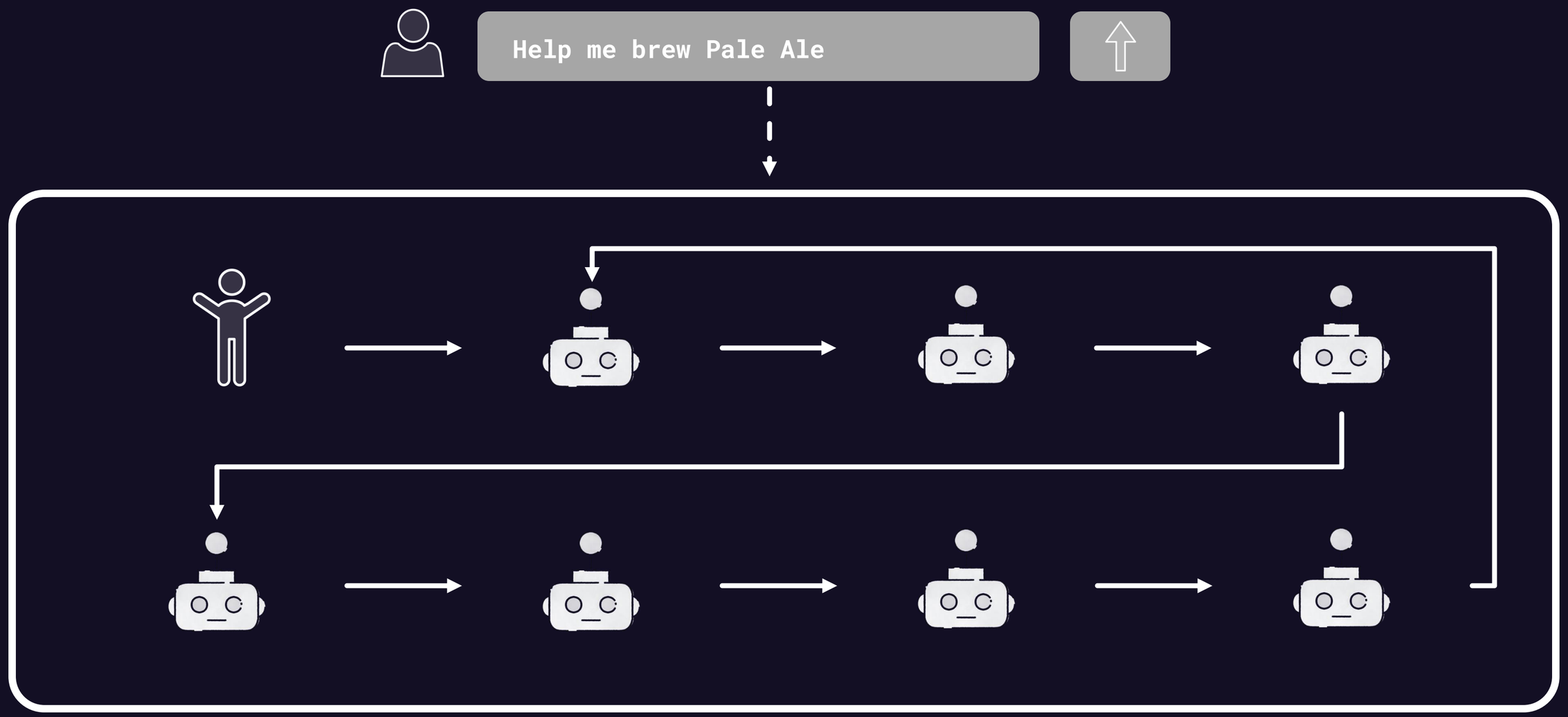

Help me brew Pale Ale

LangChain

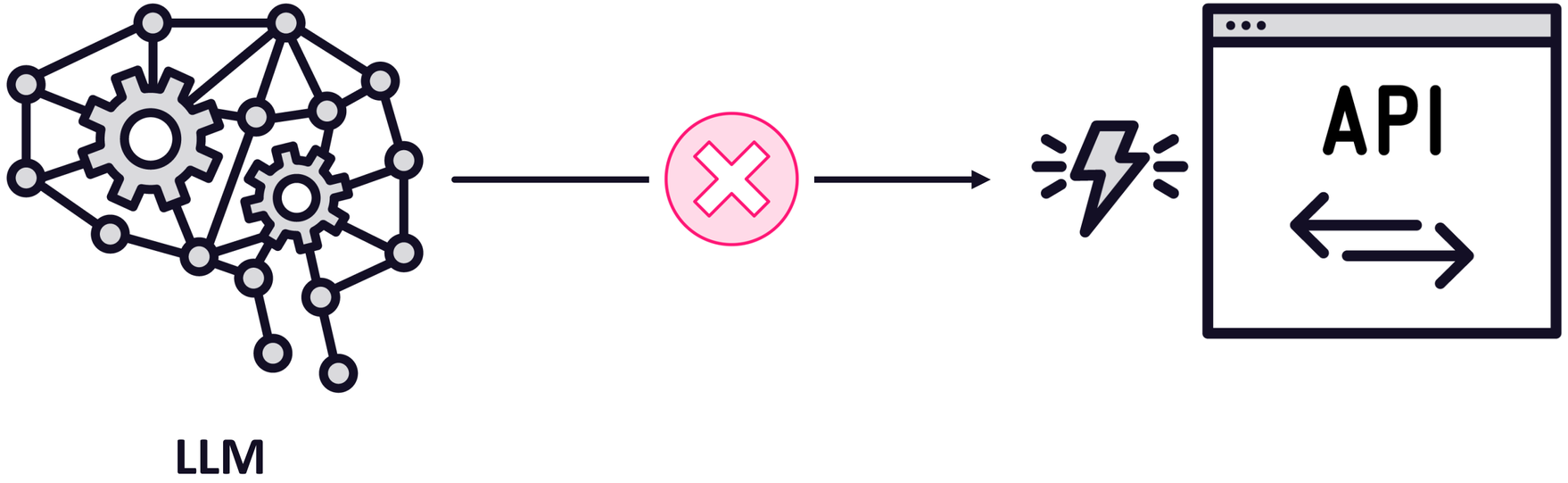
CrewAI

Microsoft AutoGen

NVIDIA Blueprints

Agents

Agents



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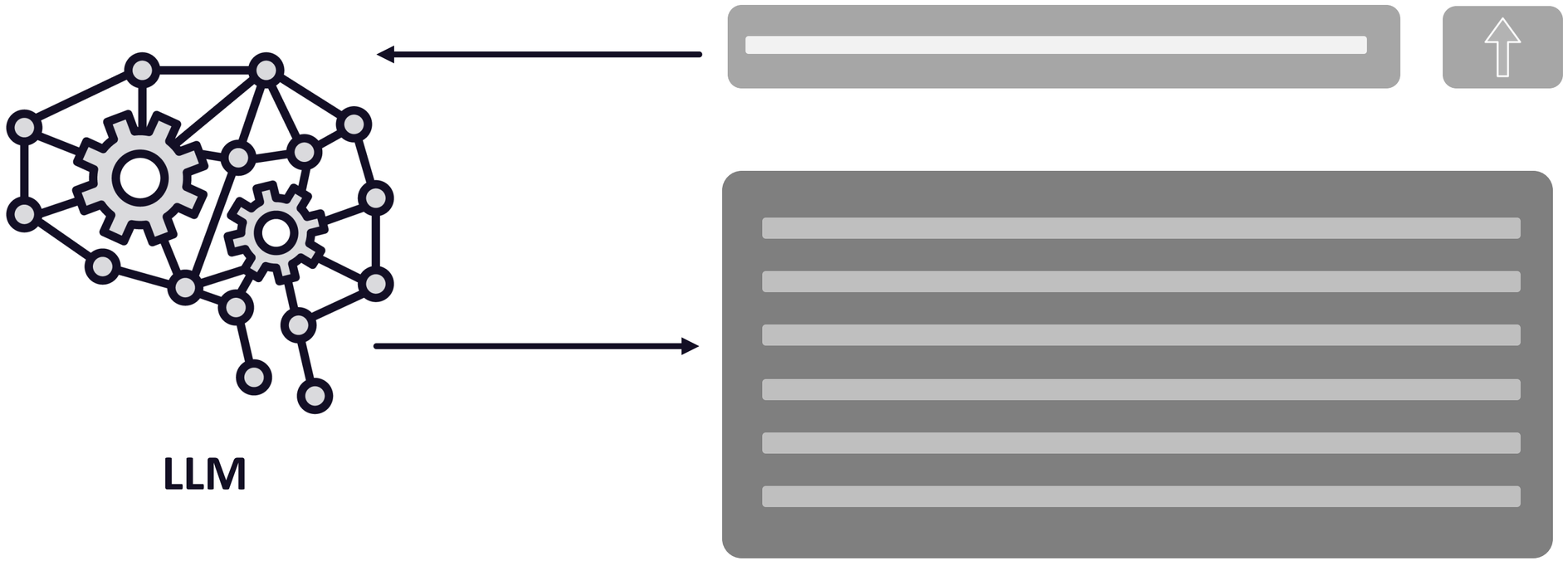

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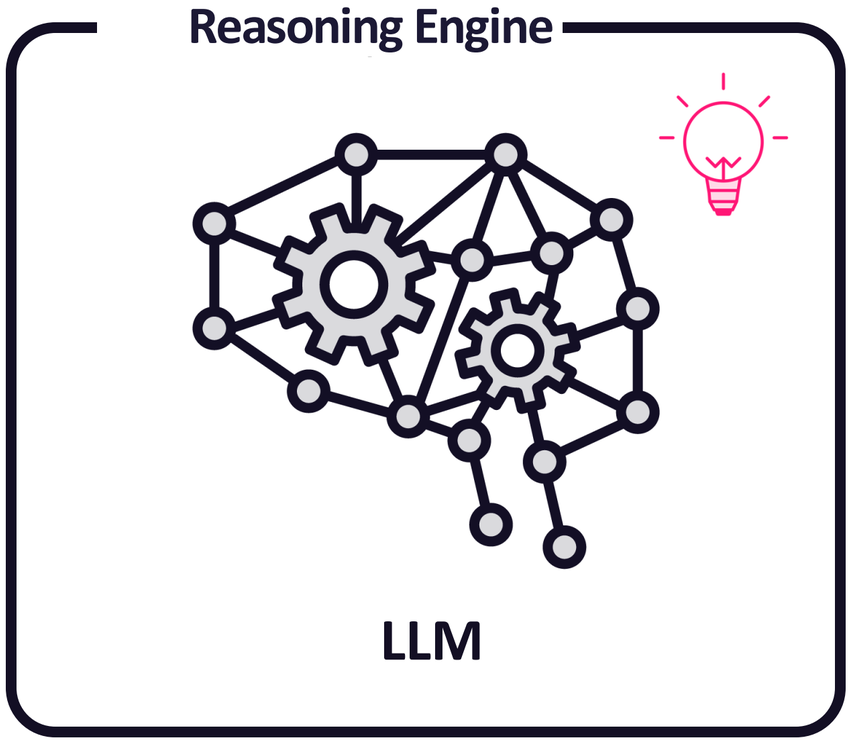
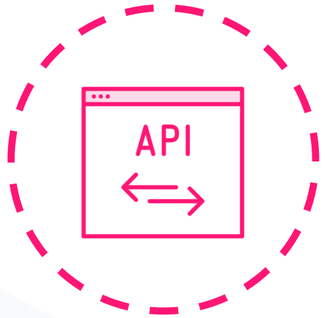


Agents

LLM

Tool executor



Agents

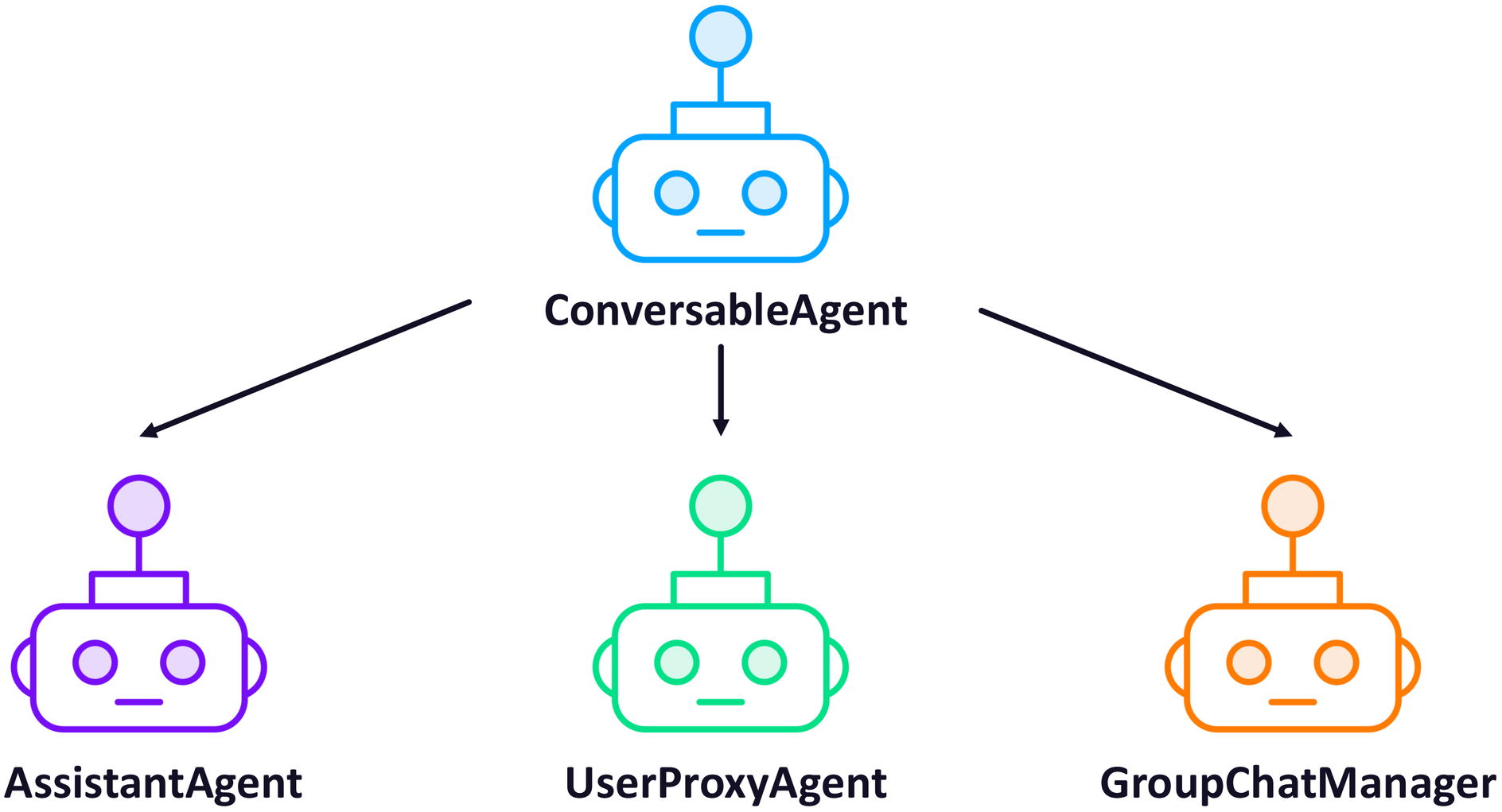
LLM

Tool executor

Code executor

Human-in-the-loop

Microsoft AutoGen Agents

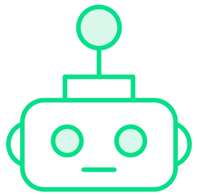


AI Agents

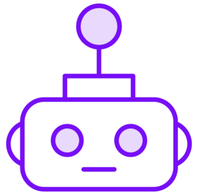
Goal

Role

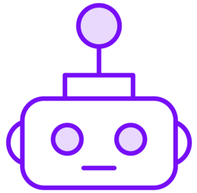
AI Agents



Type: UserProxyAgent
Role: Initiate / terminate conversation
Goal: Define initial task, provide oversight



Type: AssistantAgent
Role: Email communication specialist
Goal: Create professional, clear email responses



Type: AssistantAgent
Role: Content quality check specialist
Goal: Ensure email meets professional standards

AI Agents

Human: We need to respond to this client email about project delays.

Email Assistant: I'll draft a response addressing the delay with professionalism and clarity.
[Drafts email response]

Quality Checker: I'll review the draft for:
- Professional tone ✓
- Clear explanation of delays ✓
- Next steps outlined ✓

Suggestion: Add specific timeline for next update.

AI Agents

Quality Checker: I'll review the draft for:
- Professional tone ✓
- Clear explanation of delays ✓
- Next steps outlined ✓

Suggestion: Add specific timeline for next update.

Email Assistant: Good point. I'll add: "We will provide a detailed progress update by Friday."

Quality Checker: Perfect. The email now:
- Acknowledges the concern
- Explains the situation
- Provides clear next steps
- Sets specific expectations

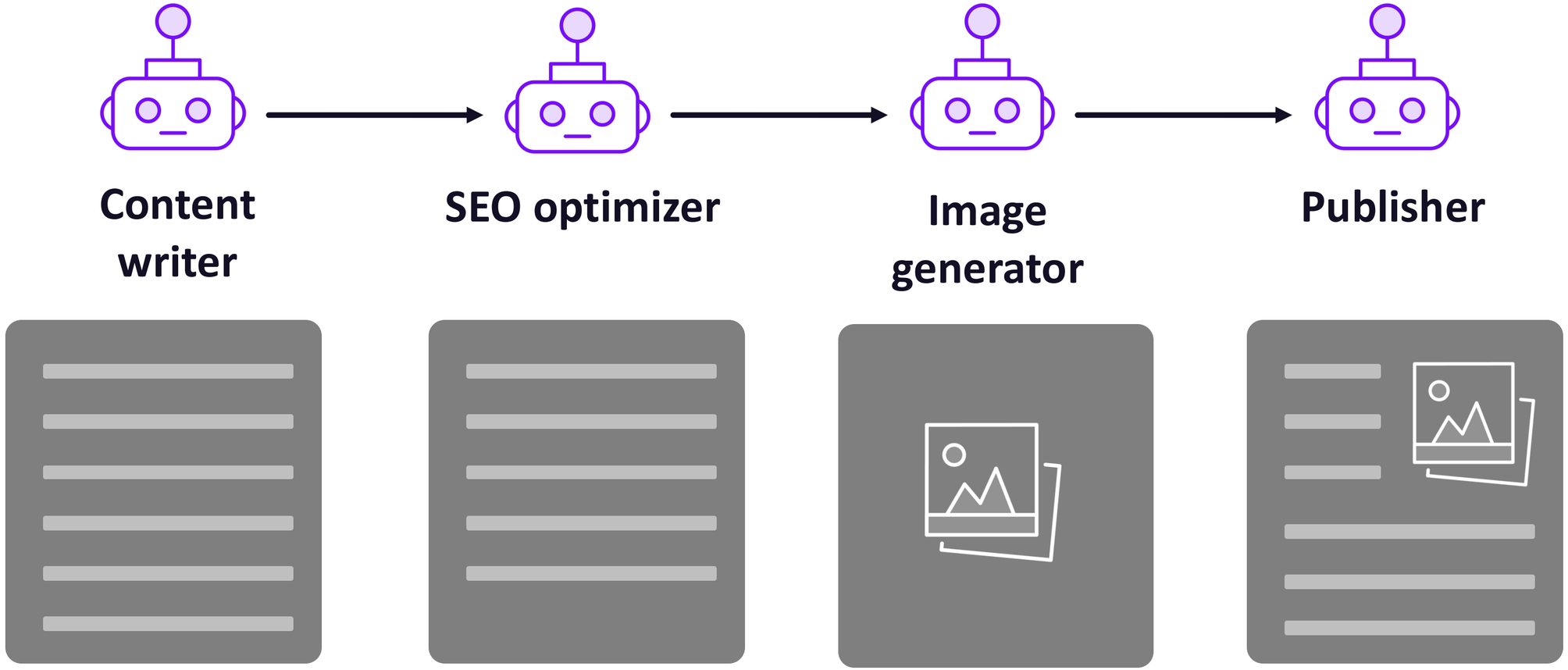
Conversational Patterns

Conversational Patterns

Two-agent chat

Sequential chat

Content Publishing



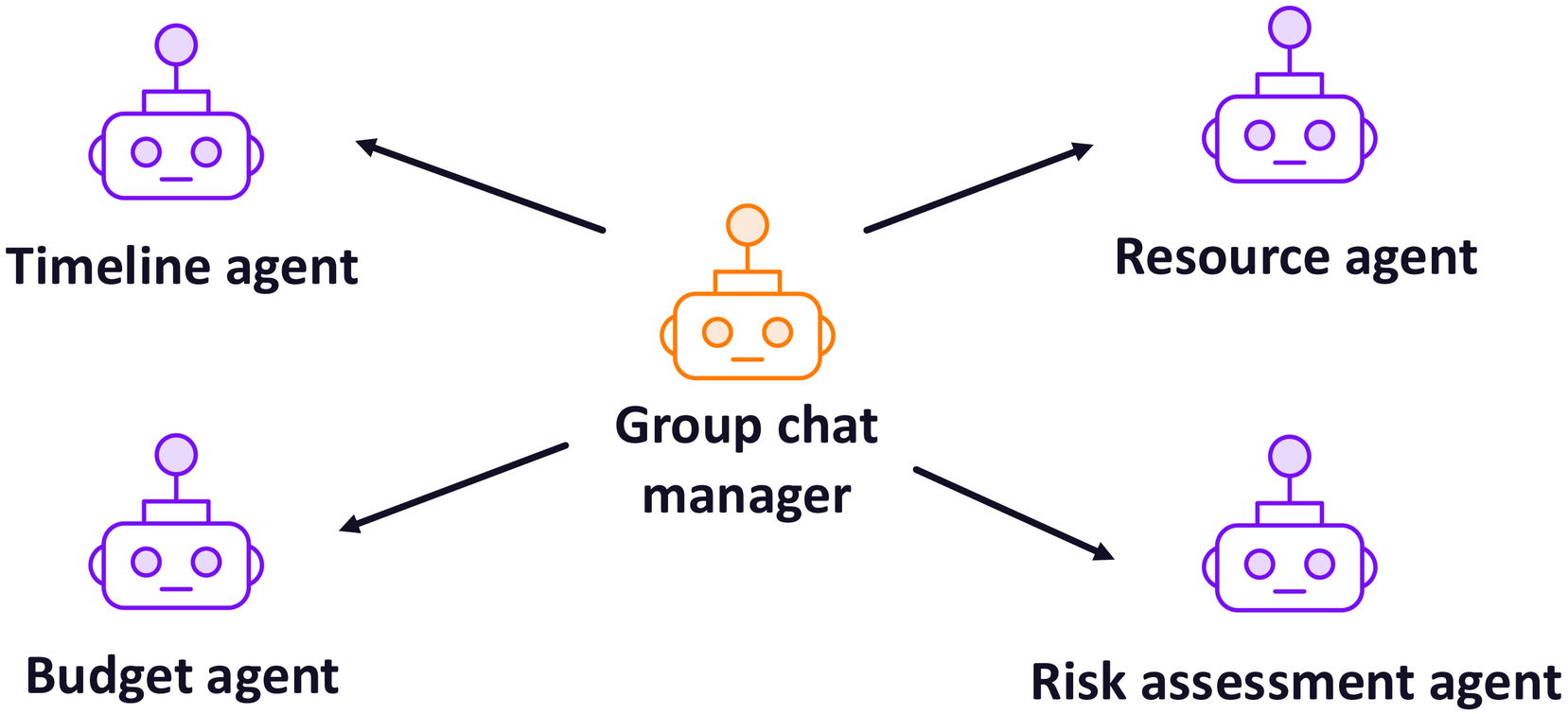
Conversational Patterns

Two-agent chat

Sequential chat

Group chat

Project Management



Conversational Patterns

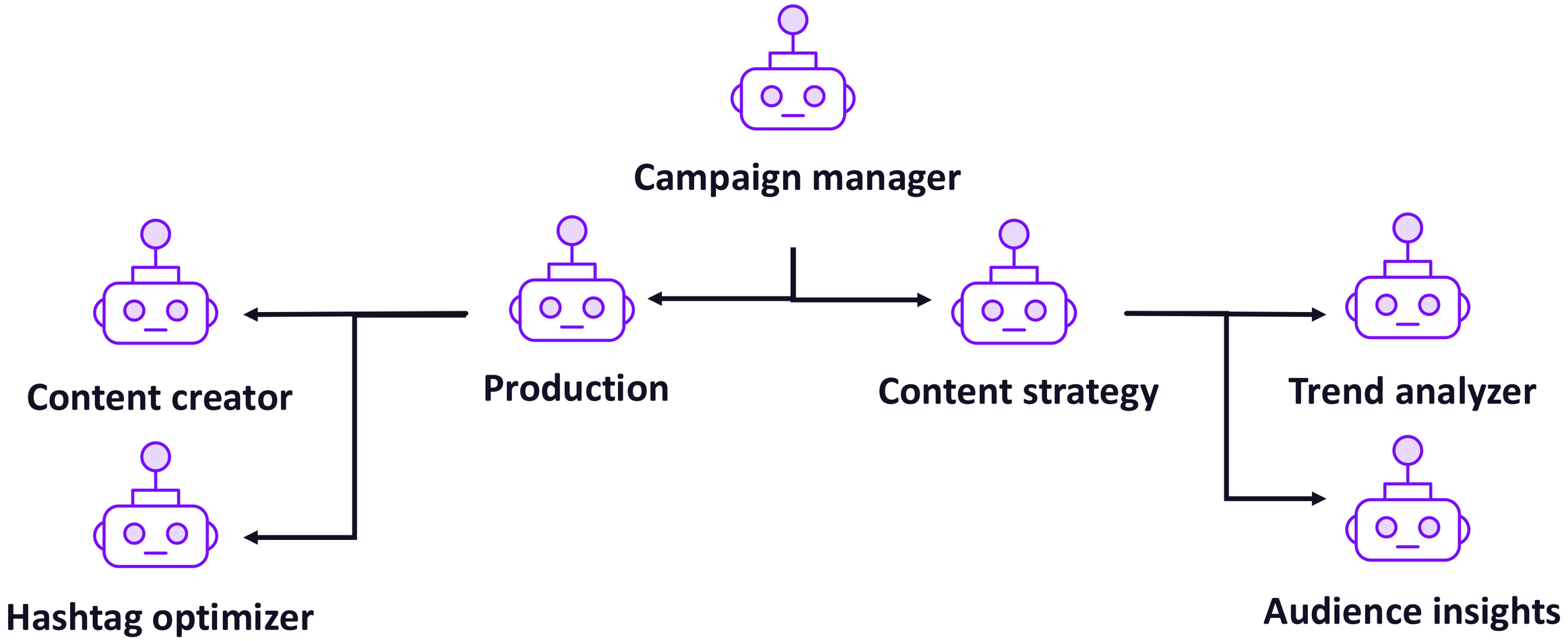
Two-agent chat

Sequential chat

Group chat

Nested chat

Social Media Campaign



Conversational Patterns

Two-agent chat

Sequential chat

Group chat

Nested chat

Demo: Text Summarizer

Thank you for your time!

LinkedIn: Bogdan Sucaciu